Deterministic Certification for the Bounded Normal Mean

An Adaptive-Support Computation with an Independent Verifier

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Abstract

We consider estimation of a normal mean in a bounded interval under squared error. A finite prior gives a lower bound on minimax risk through its Bayes risk and an upper bound through the maximum risk of its Bayes rule. We use Kempthorne's established exchange strategy to find a candidate symmetric prior and an independent outward-rounded program to verify the resulting bracket. The verifier reconstructs Gaussian-mixture ratios, bounds Simpson quadrature remainders and Gaussian tails, and covers the continuous parameter interval by branch-and-bound. It does not rely on the discovery solver's optimizer or branch tree. For the interval $[-4, 4]$, it certifies minimax risk in $[0.8150887565050, 0.8150887615302]$, with width below 5.026×10^{-9}. Posterior-flow identities provide a curvature bound for the discovery calculation; an ablation shows that its specialized common-mean enclosure yields a modest reduction in branch nodes for this instance. The certificate is a computer-assisted result conditional on the correctness of the verifier and its arithmetic implementation. We do not claim a new exchange algorithm or a finite-time convergence guarantee for the discovery procedure.

Keywords: bounded normal mean; minimax estimation; least favorable prior; interval arithmetic; branch-and-bound; computer-assisted proof.
JEL codes: C14, C44, C63. **MSC 2020:** 62C20, 65G20, 90C26.

1 Introduction

A worst-case risk guarantee for the bounded-normal mean requires a global calculation over a continuum. Let

$$Y = \theta + Z, \qquad Z \sim \mathsf{N}(0, 1), \qquad \theta \in [-m, m],$$

and use loss $(d(Y) - \theta)^2$. The objective is the minimax value

$$v^\star = \inf_d \sup_{\theta \in [-m,m]} \mathbb{E}_\theta\{d(Y) - \theta\}^2.$$

Montiel Olea and Zubova [13] recently gave a stochastic mirror-ascent method with an ex ante equally spaced parameter grid and finite-sample probability guarantees. Their paper prompted the present project. Our question is different: after a candidate finitely supported prior has been found, can a deterministic program certify its Bayes risk and the global risk of its Bayes rule without a parameter grid or Monte Carlo?

For the $m = 4$ instance, the final artifact contains three positive support levels, outward-rounded quadrature and tail enclosures, and a continuous branch-and-bound proof. A second program reconstructs the calculation from the compact certificate without importing the discovery solver.

Roadmap. Section 2 separates inherited theory, independent derivations, and implementation. Section 3 proves the exact directional derivative and minimax bracket. Section 4 develops the posterior-flow representation. Sections 5 and 6 give the continuous separator and adaptive procedure. Section 7 describes the independent verifier. Sections 8–9 report tests and compare with Montiel Olea and Zubova. The appendices supply all proof algebra, numerical error bounds, lineage details, and exact reproduction commands.

Contribution and lineage. Kempthorne [10] supplies adaptive atom insertion, joint optimization of weights and locations, pruning, the directional derivative, and the Bayes-risk/maximum-risk stopping bracket. Interval arithmetic, dependency-aware enclosures and semiconvex branch-and-bound provide the numerical foundations. This paper contributes a specialized deterministic bounded-normal implementation, a continuous separator, and an independent outward-rounded certificate. Lemma 1 gives a posterior-moment enclosure derived for this implementation; the ablation shows a four-node reduction at $m = 4$, so it is treated as a supporting device rather than the principal result.

2 Exact contribution and relation to prior work

This section states lineage before technical results become operative. Table 1 records the status used throughout the paper.

Table 1: Technical lineage and the role of each ingredient.

Object	Role	Source and exact relation
Adaptive support insertion; reoptimization of masses and locations; zero-mass or coincident-point reduction	Prior art	Kempthorne [10, pp. 175–177], Steps 1–4 and discussion.
Directional derivative and equalizer/stopping condition	Prior art	Kempthorne [10, Theorems 2.1–2.2, pp. 172–174]; the exact Gaussian algebra is supplied here for auditability.
Finite-support structure	Prior art	Kempthorne [10, Theorem 2.4, p. 175], with later structural work including [4, 5].
Deterministic global optimization of fixed finite prior parameterizations	Prior art	Gourdin, Jaumard, and MacGibbon [8, Sections 2–3].
Stochastic grid-based approximation and Wasserstein/estimator convergence	Related recent method	Theorems 1–4 of [13].
Posterior-flow identities $\mu' = V$, $V' = \kappa_3$	Standard exponential-family algebra	Re-derived in Proposition 2.
Sharp support bound for κ_3	Prior art	Equivalent to Sharma, Gupta, and Kapoor [15, Theorem 2.2 and (2.20)–(2.21)].
Common-mean product enclosure	Supporting derivation	Lemma 1; a posterior-moment enclosure with a four-node reduction at $m = 4$.
Arb verifier and $m = 4$ certificate	Release contribution	Separate direct-ratio formula, Simpson remainder, tail bound, and branch-and-bound; source and certificate accompany the paper.

Casella and Strawderman [2] establish foundational exact results for small bounds. Feldman and Brown [7] and Donoho, Liu, and MacGibbon [3] develop numerical and risk-bound calculations. Gourdin et al. [8] use theoretically convergent Lipschitz and decomposition global optimization for finite-dimensional parameterizations. Kent et al. [11] give a modified Frank–Wolfe method in probability space with oracle assumptions, while Guggenberger and Huang [9] use fictitious play and grid-based best responses in a different minimax-regret problem. Aradillas Fernández et al. [1] prove epsilon-minimax guarantees for mixtures over finitely many decision rules under a worst-case-risk oracle. The present release connects these lines of work to an outward-rounded continuous separator and an independently executable certificate for the scalar bounded-normal problem.

Interval arithmetic is classical [14]. Everitt [6] explains dependency over-enclosure and affine arithmetic in statistical computation, and Maruyama and Matsuda [12] use interval arithmetic in a computer-assisted minimaxity proof for a fixed half-Cauchy Bayes estimator. These works provide the broader dependency-aware and computer-assisted statistical context.

3 The bounded-normal minimax problem

This section defines every statistical object and proves the certificate from the original mixture formulas.

Definition 1 (Symmetric finite prior)**.** *Let $\pi = \sum_{j=1}^{J} w_j \delta_{\theta_j}$, where $\theta_j \in [-m, m]$, $w_j > 0$, and $\sum_j w_j = 1$. Symmetry means that $\pi(A) = \pi(-A)$ for every Borel set A. Define, for $r = 0, 1, 2$,*

$$f_r(y) = \sum_{j=1}^{J} w_j \theta_j^r \phi(y - \theta_j), \qquad \phi(x) = (2\pi)^{-1/2} e^{-x^2/2}.$$

Since $f_0(y) > 0$, the posterior mean and Bayes estimator are

$$\delta_\pi(y) = \frac{f_1(y)}{f_0(y)}.$$

The frequentist risk at $t \in [-m, m]$ and the Bayes risk are

$$r(t, \delta_\pi) = \int_{\mathbb{R}} \{\delta_\pi(y) - t\}^2 \phi(y - t)\, dy, \tag{1}$$

$$b(\pi) = \int_{\mathbb{R}} \left\{ f_2(y) - \frac{f_1(y)^2}{f_0(y)} \right\} dy. \tag{2}$$

Fubini's theorem gives the identity

$$b(\pi) = \sum_{j=1}^{J} w_j r(\theta_j, \delta_\pi). \tag{3}$$

The integrands are dominated because $|\delta_\pi(y)| \le m$.

Proposition 1 (Exact feasible directional derivative)**.** *For $t \in [-m, m]$ and $\pi_\alpha = (1 - \alpha)\pi + \alpha\delta_t$,*

$$\frac{d}{d\alpha} b(\pi_\alpha)\bigg|_{\alpha = 0+} = r(t, \delta_\pi) - b(\pi).$$

Proof. Write $h_r(y) = t^r \phi(y - t) - f_r(y)$, so that $f_{r,\alpha} = f_r + \alpha h_r$. Pointwise differentiation of $f_{2,\alpha} - f_{1,\alpha}^2 / f_{0,\alpha}$ gives

$$h_2 - \frac{2 f_1 h_1}{f_0} + \frac{f_1^2 h_0}{f_0^2}.$$

Substitute h_r and $\delta_\pi = f_1 / f_0$. The terms involving the original mixture equal

$$-f_2 + 2\delta_\pi f_1 - \delta_\pi^2 f_0 = -f_2 + \frac{f_1^2}{f_0},$$

whereas the atom terms equal

$$\{t^2 - 2t\delta_\pi(y) + \delta_\pi(y)^2\}\phi(y - t).$$

Integration yields $r(t, \delta_\pi) - b(\pi)$. Differentiation under the integral is justified in Appendix A; the key facts are bounded support and strict positivity of the Gaussian mixture. □

Corollary 1 (Exact minimax certificate)**.** *For every finite prior π supported on $[-m, m]$,*

$$b(\pi) \le v^\star \le U(\pi) := \sup_{t \in [-m, m]} r(t, \delta_\pi),$$

and therefore

$$0 \le v^\star - b(\pi) \le U(\pi) - b(\pi) = \sup_{t \in [-m, m]} \{r(t, \delta_\pi) - b(\pi)\}.$$

Proof. For any decision rule d, its maximum risk is at least its π-average risk, which is at least the Bayes risk $b(\pi)$. Taking the infimum over d proves the lower bound. The Bayes rule δ_π is an admissible competitor in the minimax infimum, which proves the upper bound. Subtraction and Proposition 1 give the last display. □

Corollary 1, essentially Kempthorne's stopping condition, is the logical core of the computation: a lower enclosure for $b(\pi)$ and an upper enclosure for $U(\pi)$ produce a rigorous interval for $v^\star$.

4 Posterior-flow representation

The separator must control a ratio of Gaussian mixtures without replacing its shared numerator and denominator by independent intervals. This section identifies a small posterior state that retains the relevant dependence.

For an observation y, let

$$p_j(y) = \frac{w_j \exp(\theta_j y - \theta_j^2/2)}{\sum_k w_k \exp(\theta_k y - \theta_k^2/2)}.$$

The omitted common factor $\phi(y)$ cancels. Define the posterior mean, variance, and cumulants

$$\mu(y) = \sum_j p_j(y)\theta_j, \quad V(y) = \sum_j p_j(y)\{\theta_j - \mu(y)\}^2, \quad \kappa_s(y) = s\text{th posterior cumulant.}$$

Proposition 2 (Posterior flow and risk curvature)**.** *For a finite bounded prior, all functions above are analytic and*

$$\mu' = V, \qquad V' = \kappa_3, \qquad \kappa_s' = \kappa_{s+1}.$$

With $q(t,z) = \mu(t+z) - t$ and $Z \sim \mathsf{N}(0,1)$,

$$r(t,\delta_\pi) = \mathbb{E}q(t,Z)^2, \tag{4}$$
$$r'(t,\delta_\pi) = 2\mathbb{E}\{q(t,Z)(V(t+Z)-1)\}, \tag{5}$$
$$r''(t,\delta_\pi) = 2\mathbb{E}\{(V(t+Z)-1)^2 + q(t,Z)\kappa_3(t+Z)\}. \tag{6}$$

Proof. The probabilities form a finite natural exponential family with natural parameter y. Successive derivatives of

$$\log\left(\sum_j w_j e^{\theta_j y - \theta_j^2/2}\right)$$

produce the cumulants. Further, $\partial_t q = V(t+z) - 1$ and $\partial_t V = \kappa_3$. Two differentiations of $\mathbb{E}q^2$ give the displayed formulas. Bounded support supplies an integrable uniform bound for each derivative. □

The only sign-indefinite product in (6) is $q\kappa_3$. The following representation couples its factors.

Lemma 1 (Common-mean representation and enclosure)**.** *Let the posterior be supported on $[\alpha,\beta]$. If $V > 0$, define*

$$\nu = \frac{\mathbb{E}\{\Theta(\Theta-\mu)^2 \mid y\}}{V}.$$

Then $\nu \in [\alpha,\beta]$ and

$$\kappa_3 = V(\nu - \mu).$$

If $t \in [a,b]$, $\mu \in M = [\mu_L, \mu_U]$, and $0 \le V \le V_U$, then

$$q\kappa_3 \le V_U \max_{\mu\in M}\big\{(\mu-a)_+(\beta-\mu),\, (b-\mu)_+(\mu-\alpha)\big\}, \tag{7}$$
$$q\kappa_3 \ge -V_U \max_{\mu\in M}\big\{(\mu-a)_+(\mu-\alpha),\, (b-\mu)_+(\beta-\mu)\big\}. \tag{8}$$

Each maximum is either an endpoint value or the value at the vertex of one quadratic.

Proof. The weights proportional to $p_j(\theta_j-\mu)^2$ are nonnegative and sum to V, hence ν is their mean and lies in the convex hull $[\alpha,\beta]$. Expanding $\mathbb{E}\{\Theta(\Theta-\mu)^2\} = \kappa_3 + \mu V$ proves the identity. For a positive product, if $\mu \ge t$ choose the largest feasible $\nu = \beta$; if $\mu \le t$ choose $\nu = \alpha$. Maximizing over $t \in [a,b]$ gives (7). The opposite choices of ν give the most negative product and (8). The case $V = 0$ has $\kappa_3 = 0$ and follows by continuity. □

Equality and degeneracy. Equality in the identity holds by construction. An envelope endpoint can be attained only when the posterior squared-deviation weights put their mean ν at the relevant support endpoint and when $V = V_U$; $V = 0$ collapses all terms to zero. These conditions are useful diagnostics but are not assumed by the solver.

Scope of the enclosure. The relaxation separates V from μ through V_U and encloses, rather than identifies, their exact joint range. The ablation compares it with a Cartesian product of independent intervals for q and κ_3. The results in Section 8 measures the remaining over-enclosure.

5 Certified continuous separation

This section turns point-risk and curvature enclosures into a global upper bound. The final verifier uses direct mixture-ratio interval boxes rather than the specialized common-mean formula, providing formula-level independence.

Lemma 2 (Semiconvex secant majorant)**.** *Let f be twice differentiable on $[a,b]$ and $f'' \geq -K$ there, with $K \geq 0$. For $x = a + s(b-a)$, $s \in [0,1]$,*

$$f(x) \leq (1-s)f(a) + sf(b) + \frac{K}{2}(b-a)^2 s(1-s).$$

Consequently, the cell maximum is bounded by the maximum of the two endpoints and, when its vertex lies in $[0,1]$, the vertex value of the quadratic on the right.

Proof. The function $f(x) + Kx^2/2$ is convex. Apply the defining secant inequality for a convex function and collect the quadratic terms. No assumption about the sign of f' is required. □

Proposition 3 (Risk quadrature enclosure)**.** *Fix $t \in [-m,m]$ and $A > 0$. Composite Simpson quadrature on $[-A,A]$, with an interval enclosure of the fourth derivative on every panel, encloses the truncated risk integral. The omitted tails satisfy*

$$0 \leq \int_{|z|>A} q(t,z)^2 \phi(z)\, dz \leq 8m^2 \frac{\phi(A)}{A}.$$

Proof. The panel error is $-h^5 H^{(4)}(\xi)/2880$ when h denotes the full panel width. Replacing $H^{(4)}(\xi)$ by its interval range is inclusion monotone. Since $|\mu - t| \leq 2m$, the tail is at most $4m^2 \Pr(|Z| > A)$; Mills' inequality $\Pr(|Z| > A) \leq 2\phi(A)/A$ gives the result. □

The verifier obtains a lower enclosure for (6) by partitioning $[-A,A]$ in z, evaluating the common posterior ratio on the box $t \in [a,b]$, and adding a two-sided curvature-tail enclosure. If $0 \leq V \leq m^2$, then

$$|(V-1)^2 + q\kappa_3| \leq \max\{1, (m^2-1)^2\} + 4m^4,$$

because $|q| \leq 2m$ and $|\kappa_3| \leq 2mV \leq 2m^3$. This gives an explicit tail bound for (6).

Theorem 1 (Verifier implication)**.** *Suppose an outward-rounded computation returns bounds satisfying*

$$B_L \leq b(\pi) \quad \text{and} \quad U_U \geq \sup_{t \in [-m,m]} r(t, \delta_\pi).$$

Then

$$v^\star \in [B_L, U_U], \qquad 0 \leq v^\star - B_L \leq U_U - B_L.$$

If $U_U - B_L \leq \varepsilon$, the Bayes rule is certifiably ε-minimax in value.

Proof. This is Corollary 1 with interval endpoints weakened outward. It turns the established minimax bracket into an executable certification rule. □

6 Adaptive computational procedure

The outer loop below is Kempthorne's algorithm specialized to symmetric bounded-normal priors. Symmetry reduces the search to nonnegative levels; a positive level represents an equal pair at $\pm t$, while level zero represents one atom.

Adaptive support computation followed by certification. Given $m > 0$, a requested gap $\varepsilon > 0$, and an initial symmetric prior:

1. reoptimize retained level masses;
2. optionally optimize interior support locations while keeping them ordered;
3. prune zero masses and merge numerically coincident levels;
4. compute a continuous global risk maximizer or upper bound on $[0, m]$;
5. if the numerical separation gap is not small, add the symmetric pair at the maximizing level and return to step 1;
6. serialize only m, levels, masses, verifier settings, and claims;
7. run the independent outward-rounded verifier; and
8. return the prior, Bayes rule, $[B_L, U_U]$, and report only if the verifier proves $U_U - B_L \le \varepsilon$. Otherwise refine and repeat.

Figure 1: The outer loop is inherited from Kempthorne; certification is a separate post-processing stage.

Fast mode. The discovery solver uses float64, deterministic Gauss–Hermite quadrature, warm starts, quasi-Newton-style mass and location updates, aggressive pruning, and the common-mean curvature enclosure. Its bounds are diagnostics, not certificates. No parameter grid is fixed in advance; branch cells are created adaptively.

Certificate mode. The final computation uses 192-bit Arb balls. The verifier recomputes posterior ratios, Simpson sums and remainder intervals, tail bounds, cell curvatures, and continuous branch coverage. It accepts no discovery-solver branch tree.

Convergence statement. Kempthorne [10, Theorem 3.1] proves asymptotic convergence under assumptions A1–A5 for the abstract exchange procedure. The executable guarantee here is conditional and post hoc: whenever the independent verifier accepts, Theorem 1 proves the reported gap. A finite iteration bound for finding a certifiable prior remains open.

7 Independent certificate and verifier

The compact certificate contains m, nonnegative support levels, total level masses, arithmetic precision, quadrature and branch settings, and claimed risk intervals. It contains no estimator samples, parameter grid, Monte Carlo draws, or trusted branch partition.

The verifier performs the following checks.

1. It rejects nonfinite input, support outside $[0, m]$, unordered or duplicate levels, nonpositive masses, and masses that do not sum exactly to one as decimal inputs.
2. It reconstructs both atoms at every positive level and computes the posterior directly from $w_j \exp(\theta_j y - \theta_j^2/2)$.
3. It evaluates the risks at all support levels with independent composite Simpson code, fourth-derivative intervals, and Gaussian tails, then averages them to enclose the Bayes risk.
4. It starts from the complete interval $[0, m]$. Each branch cell receives a curvature lower bound and the exact semiconvex quadratic majorant. Only a cell whose upper bound is within the work tolerance of the incumbent may be pruned.
5. It checks that the supplied claim intervals contain the recomputation and that the final gap is below the requested tolerance.

The verifier imports neither `solver_fast.py` nor `solver_certified.py`. It does share the mathematical specification: Gaussian density, posterior mean, risk, and the semiconvex theorem. Complete formula independence would be impossible without changing the quantity being verified. The implementation of those formulas and the arithmetic library are independent.

Trusted computing base. The final proof trusts CPython, python-flint 0.8.0, the Arb/FLINT implementation beneath it, and the verifier source. The discovery optimizer and NumPy are outside the final trusted base. The resulting assurance level is a computer-assisted proof with an explicit software trusted base.

7.1 AI assistance

AI tools assisted with literature search, theorem exploration, drafting, code generation and adversarial testing in the project. During preparation of this revision, OpenAI GPT-6.1 Terra assisted with manuscript revision and GPT-6.1 Sol assisted with code and build work. AI-generated suggestions are not mathematical evidence. The verifier and its arithmetic library form part of the trusted computing base; an independently executable program is not an independent human review.

8 Numerical results and ablations

All timings in this section are single-process measurements on the local macOS 15.7.3 arm64 machine and are descriptive rather than hardware-normalized complexity claims.

8.1 The certified result for m=4

The final prior has nonnegative levels

$$(0.5984699814893856,\ 2.1264117002421385,\ 4)$$

and total level masses

$$(0.36699366016033996,\ 0.3994285510645993,\ 0.23357778877506074).$$

Each positive level is split equally between its positive and negative atom. The independent 192-bit verification used tail cutoff $A = 8$, 2048 Simpson panels per point risk, 512 curvature cells, and 567 branch nodes. It took 49.68 seconds in the recorded clean run.

Table 2: Outward-rounded $m = 4$ certificate. Displayed endpoints are rounded outward from the longer Arb output.

Quantity	Independently recomputed enclosure
Bayes risk	[0.8150887565050, 0.8150887565548]
Worst-case risk	[0.8150887565301, 0.8150887615302]
Minimax risk	[0.8150887565050, 0.8150887615302]
Certified gap upper bound	$5.02517E-9$
Branch nodes	567
Arithmetic	python-flint Arb balls, 192 bits

The verifier's Bayes-risk width is 4.98×10^{-11}. The branch envelope uses a 5.0×10^{-9} work width. The certified minimax-gap upper bound is 5.02517×10^{-9}, below 10^{-8}.

8.2 Fast deterministic sweep

Table 3 reports float64 discovery results. They are not certificates except where a separate certificate is stated.

Table 3: Adaptive-support float64 sweep; no ex ante parameter grid.

m	levels	atoms	Bayes risk	float gap	seconds
0.25	1	2	0.0588158765	0.00e+00	0.003
0.5	1	2	0.1989864336	0.00e+00	0.000
1	1	2	0.4495995092	0.00e+00	0.001
1.5	2	3	0.5561720123	3.78e-10	0.020
2	2	3	0.6448008729	3.47e-09	0.030
3	3	5	0.7506254421	3.62e-09	0.076
4	3	6	0.8150887565	2.64e-09	0.187
5	4	7	0.8573009664	3.71e-09	0.218
8	6	11	0.9231345345	6.26e-09	5.149

8.3 Dependency ablation

Table 4 holds the prior and endpoint-risk evaluator fixed. The two generic global bounds exhaust the 5001-node budget. Natural interval, one-noise affine, Cartesian posterior-flow, and common-mean flow bounds terminate. These rows are float64 diagnostics; the Arb calculation supplies the certified result.

Table 4: $m = 4$, target separator width 10^{-8}.

Separator	nodes	exhausted	width	seconds
global Lipschitz	5001	yes	2.34e-01	0.115
global semiconvex	5001	yes	5.09e-04	0.113
natural interval	291	no	8.58e-09	2.248
one-noise affine	427	no	9.95e-09	2.208
Cartesian flow	281	no	5.78e-09	3.170
common-mean flow	277	no	4.57e-09	0.076

The common-mean formula reduces the Cartesian-flow tree from 281 to 277 nodes. Its much shorter runtime in this implementation comes primarily from vectorized engineering. Across cell widths from approximately 3.87 to 10^{-5}, the ratio of common-mean to Cartesian curvature width ranged from 0.864 to 0.996. The measured mathematical tightening is therefore modest.

8.4 Numerical checks

A seeded set of 100 valid symmetric priors and parameter intervals produced no dense-sample violation of the float64 curvature enclosure. Such checks provide falsification evidence; outward-rounded inclusion supplies the proof.

9 Comparison with Montiel Olea and Zubova

Montiel Olea and Zubova [13] use stochastic mirror ascent on an equally spaced grid. Their theory controls discretization and stochastic optimization error, proves Wasserstein convergence of approximate priors, and proves convergence of the resulting estimators. The method is expressed through broadly applicable first-order stochastic optimization. The present solver exploits scalar Gaussian posterior identities to produce a deterministic certificate for this bounded-normal instance.

For a numerical cross-check, we reconstructed their stated tuning formulas and digitized the published risk figure using a checksum-locked pixel calibration. The deterministic values for $m = 1, 2, 3, 4$ lie within the displayed figure intervals to the declared pixel error.

At $m = 4$ the present float64 discovery solver took 0.25 seconds in the clean reproduction, and the independent certificate took 49.68 seconds. These times describe the current implementation on the recorded machine. The two methods target different computational guarantees, so the comparison is made through their outputs and assurance levels rather than a speed ratio.

The deterministic certificate independently validates the bounded-normal minimax value used in the local-projection/vector-autoregression application of Montiel Olea and Zubova. Its scope is the scalar minimax ingre-

Table 5: Complementary computational objectives.

Feature	Montiel Olea–Zubova	Present project
Parameter representation	Fixed equally spaced grid	Adaptive continuous branch cells; finite active support
Optimization	Stochastic mirror ascent over grid weights	Kempthorne exchange; float64 mass/location reoptimization
Integration	Monte Carlo supergradient and reported risk evaluation	Deterministic Gauss–Hermite discovery; interval Simpson verification
Guarantee	High-probability approximation and convergence theory	Post-hoc deterministic value certificate for each accepted prior
Continuous separation	Lipschitz grid correction	Outward-rounded branch-and-bound on $[0, m]$
Generality	First-order method with broader optimization relevance	Specialized scalar bounded-normal implementation

dient. The stochastic method retains the broader optimization and estimator-convergence theory established in their paper.

10 Scope and conclusion

The final certificate concerns a one-dimensional normal location model with unit variance, squared loss, bounded symmetric parameter space, and a supplied finite symmetric prior. Kempthorne's convergence theorem governs the abstract exchange procedure; the executable guarantee here is post hoc: acceptance by the independent verifier proves the reported minimax bracket. The specialized branch tree has no finite worst-case complexity bound in this release. Its trusted base is CPython, python-flint, Arb/FLINT and the verifier source. The $m = 4$ result receives the final Arb certificate; the remaining sweep records float64 discovery evidence.

The paper combines established least-favorable-prior optimization with a continuous separator and independent outward-rounded verification. The common-mean enclosure gives a modest node reduction in the reported ablation; the compact executable certificate is the main deliverable. For $m = 4$ it proves a minimax-risk interval of width below 5.026×10^{-9} and exposes every numerical tolerance required to replay that statement.

Declarations

Code and reproducibility. The implementation, final certificate, lockfile and reproduction commands are available at github.com/Maar10Herr/Bounded-Normal-Minimax-Certificate-Public.

Software and hardware. The recorded environment used CPython 3.13.5, NumPy 2.3.2, Pillow 11.3.0 and python-flint 0.8.0 on macOS 15.7.3 arm64. Reported timings are descriptive machine measurements; the certified inequalities follow from the outward- rounded verifier.

Acknowledgements. The project was motivated by the bounded-normal minimax work of José Luis Montiel Olea and Ekaterina Zubova; their work and the classical literature are cited throughout.

A Proof details

A.1 Differentiation of Bayes risk

Fix $\alpha_0 < 1$. For $\alpha \in [0, \alpha_0]$, $f_{0,\alpha} \geq (1 - \alpha_0) f_0$. Directly expanding the differentiated integrand in Proposition 1 reduces it to the sum of $-f_2 + f_1^2/f_0$ and $(t - \delta_\pi)^2 \phi(\cdot - t)$. The first absolute value is bounded by $f_2 + m^2 f_0$, and the second by $4m^2\phi(\cdot - t)$. Both are integrable. Dominated convergence therefore justifies the right derivative. The same argument applies to a direction given by any bounded-support probability measure after integrating the atom formula in t.

A.2 Fourth derivative used by Simpson's rule

Let $u(z) = q(t,z)^2$. With derivatives taken in z,

$$\begin{aligned} u_0 &= q^2, \\ u_1 &= 2qV, \\ u_2 &= 2(V^2 + q\kappa_3), \\ u_3 &= 2(3V\kappa_3 + q\kappa_4), \\ u_4 &= 2(3\kappa_3^2 + 4V\kappa_4 + q\kappa_5). \end{aligned}$$

Because $\phi' = -z\phi$, repeated product differentiation gives

$$(u\phi)^{(4)} = \phi\big[u_4 - 4zu_3 + 6(z^2-1)u_2 + 4(3z - z^3)u_1 + (z^4 - 6z^2 + 3)u_0\big].$$

Every posterior quantity is evaluated on the same (t,z) box.

A.3 Curvature tail

For posterior support in $[-m,m]$, $0 \le V \le m^2$ by the Bhatia–Davis inequality. Also

$$|\kappa_3| = \big|\mathbb{E}(\Theta-\mu)^3\big| \le 2m\,\mathbb{E}(\Theta-\mu)^2 = 2mV.$$

Thus the absolute curvature state is no greater than

$$C_m = \max\{1, (m^2-1)^2\} + 4m^4.$$

Outside $[-A,A]$, the magnitude omitted from (6) is at most $2C_m \Pr(|Z| > A) \le 4C_m\phi(A)/A$.

A.4 Branch pruning invariant

At every time the active heap plus already pruned cells covers $[0,m]$. An active cell has a proved upper bound. A cell is pruned only if its bound is no larger than the current incumbent plus the work tolerance. Therefore the global supremum is bounded by the maximum of the active-cell bounds and the incumbent plus the work tolerance. This maximum is exactly the value reported by the verifier. The use of floating-point heap priorities affects only processing order; each retained mathematical bound is an Arb ball.

B Implementation and verification details

B.1 Error budget at m=4

The independently recomputed Bayes-risk enclosure width is 4.9777×10^{-11}. Of this, the reported Simpson-remainder width is 4.9696×10^{-11}, the Gaussian tail upper bound is 8.084×10^{-14}, and arithmetic rounding in the Simpson sum is below 3.1×10^{-40}. At risk points encountered by separation, the largest fourth-order remainder width is 6.5690×10^{-11} and the largest tail upper bound is 8.084×10^{-14}. The branch envelope width is 5.0×10^{-9}.

B.2 Independent reconstruction

The discovery solver computes posterior states with vectorized NumPy and a specialized flow tube. The final verifier instead:

1. constructs Arb balls from decimal strings;
2. evaluates each mixture exponential directly;
3. forms posterior central moments through order five;
4. implements a separate Simpson panel loop;

5. uses direct interval boxes for curvature rather than the common-mean flow formula; and

6. rebuilds its own branch tree from $[0, m]$.

The two paths share the mathematical definition of risk while using different numerical representations and control flow.

B.3 Failure modes examined

Independent intervals for f_0, f_1^2, and $1/f_0$ repeatedly count the same uncertainty and become particularly wide in the tails. A global Lipschitz bound scales only linearly with cell width and exhausted the 5001-node budget even at tolerance 10^{-4}. A global curvature bound also exhausted that budget. One-noise affine arithmetic helps on narrow cells but was sometimes wider than the posterior-flow representation. These empirical results characterize the tested representations and tolerances.

C Detailed prior-art comparison

Kempthorne's Section 3 begins with a finite prior, locally maximizes Bayes risk over weights and locations, globally maximizes the Bayes-rule risk, stops when maximum risk equals Bayes risk, and otherwise inserts the maximizer [10, pp. 175–177]. He also notes that zero weights or coincident locations can reduce support. His implementation brackets local risk maxima by a parameter grid and refines them by golden-section search [10, p. 177]. The present verifier adds outward rounding and complete continuous branch coverage to that optimization lineage.

Gourdin et al. terminate their Lipschitz global optimizer when the gap between an upper bound and incumbent falls below a chosen tolerance [8, pp. 19–21]. Their bounded-normal discussion fixes a finite-dimensional prior form justified for a small range of m [8, pp. 25–26]. The present solver combines adaptive support generation with an interval-certified continuous risk oracle.

Dytso et al. [5] give support-cardinality reductions under a finite observation alphabet and use projected gradient ascent in numerical examples. Their stated finite-alphabet theorem addresses a different observation space from the scalar Gaussian model. Kent et al. [11] assume derivative and local linear-minimization oracles and use stochastic subroutines. Guggenberger and Huang [9] obtain iterationwise brackets in a finite/discretized fictitious-play problem with grid-based best responses. These results locate the current work within broader finite-mixture and minimax-optimization methods.

Maruyama and Matsuda [12] provide the closest located interval-assisted normal-mean minimaxity precedent. They verify inequalities for a Bayes estimator under a fixed half-Cauchy prior in a multivariate problem. The certified object here is instead the Bayes/maximum-risk gap for a least-favorable prior in the bounded scalar problem.

This comparison is source-grounded and records the closest antecedents used in the paper. Future revisions should incorporate any closer continuous, outward-rounded bounded-normal certificate brought to the author's attention.

D Reproducibility

The repository linked in the declarations contains the verifier, certificate, lockfile and setup instructions. From its root, the certificate can be replayed with

```
uv run --frozen python src/verification/verify_certificate_arb.py \
  certificates/final/m4_epsilon_1e-8_arb.json
```

The verification reconstructs the Bayes-risk lower bound and the global risk upper bound for the specified finite symmetric prior. Its conclusion is conditional on the trusted computational base described in Section 7.

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